

50 Cities to visit before you classify - A world-wide multi-modal dataset for semantic segmentation of remote sensing imagery

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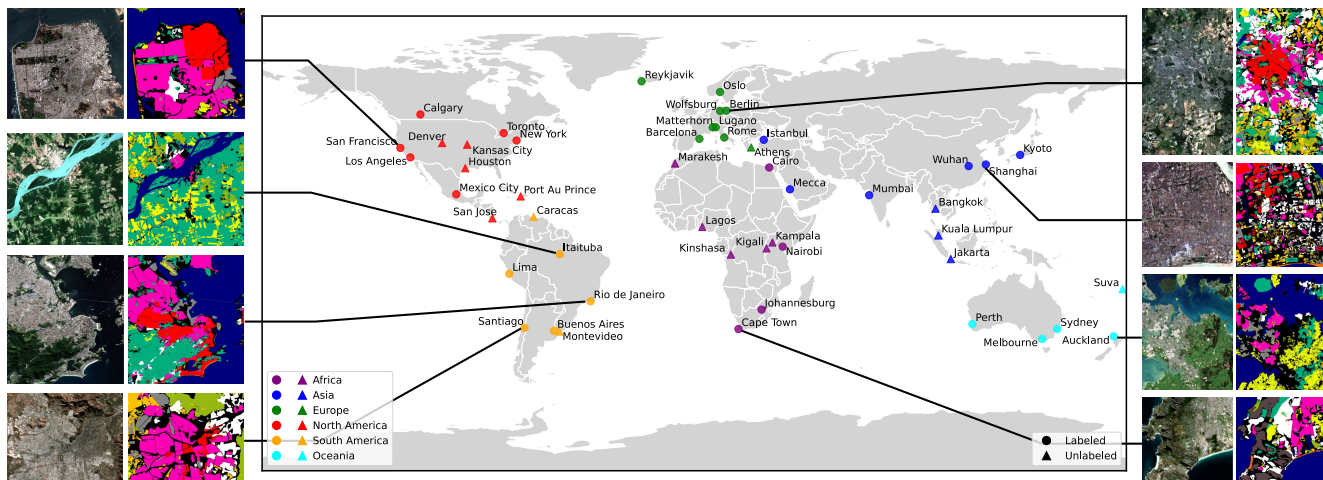


Figure 1. We propose the 50 Cities dataset, a global-scale benchmark for multi-modal land-cover segmentation across 50 urban regions on six continents, including 33 labeled and 17 unlabeled cities. The map illustrates the global distribution of all cities. Example RGB Sentinel-2 scenes and corresponding semantic maps highlight the diversity of urban structures and surrounding landscapes.

Abstract

Earth observation imagery is vital for land use monitoring, urban planning, and environmental management. Semantic segmentation of satellite data enables global Land Use and Land Cover (LULC) mapping, yet progress is limited by the scarcity of accurate benchmarks as existing datasets often rely on semi-automatically generated labels, patch-based sampling, and local geographic regions. We present 50 Cities, a new benchmark for multi-modal LULC segmentation and cross-continental generalization. It offers multi-modal satellite imagery across 50 urban regions worldwide, including 33 cities with manual pixel-level annotations and 17 unlabeled cities for semi- and self-supervised learning. Each city is provided as a large, continuous scene capturing diverse urban morphologies and surrounding landscapes. Comprehensive experiments with five baselines, ranging from a simple U-Net to a foundation model, establish strong references and highlight the challenges of global multimodal segmentation. The dataset, code, and models are available at github.com/rhaensch/50cities.

1. Introduction

Earth observation (EO) plays a crucial role in addressing global challenges such as climate change, sustainable urbanization, agriculture, and disaster response [17]. By providing consistent, high-resolution imagery of the Earth’s surface, EO enables data-driven decision-making at a planetary scale. A central application is Land Use and Land Cover (LULC) mapping, which assigns semantic labels to pixels in satellite or aerial imagery to describe human and natural environments. The increasing availability of freely accessible satellite data has fueled rapid progress in machine learning methods for large-scale LULC mapping.

Driven by this data abundance, numerous benchmark datasets have emerged for semantic segmentation of EO imagery. These range from optical-only datasets such as LandCoverNet [1] and Dynamic World [5] to multimodal benchmarks like SEN12MS [21] and reBEN [7], and more recent large-scale resources such as MMEarth [16] and TerraMesh [3] designed for geospatial foundation model train-

ing. Despite their impact, existing datasets often suffer from limited geographic diversity, reliance on coarse or automatically derived reference labels, and a patch-based structure that restricts spatial context. These factors constrain achievable accuracy and hinder research on cross-continental generalization in heterogeneous global environments.

To address these challenges, we introduce *50 Cities* (illustrated in Fig. 1), a new benchmark for multimodal LULC segmentation and domain generalization. It provides co-registered Sentinel-1 (SAR) and Sentinel-2 (MSI) imagery at 10 m resolution across 50 globally distributed metropolitan areas spanning six continents. The dataset includes manual annotations for 13 LULC classes in 33 cities and an additional 17 unlabeled scenes for semi-supervised learning. Unlike patch-based datasets, *50 Cities* offers large, continuous urban regions capturing diverse land-cover and land-use patterns.

We further present baseline experiments for supervised, semi-supervised, and self-supervised segmentation. Results highlight both the capability of current methods and the substantial potential for future advances in global multimodal LULC mapping.

The *50 Cities* dataset offers the following key properties:

- **Global coverage:** 50 metropolitan regions across six continents.
- **Accurate reference data:** Manually annotated 13-class LULC maps for 33 cities.
- **Dual-modality:** Co-registered Sentinel-1 and Sentinel-2 imagery at 10 m resolution.
- **Strong baselines:** Comprehensive experiments establishing reliable performance references.

2. Related Work

A wide variety of datasets have been developed for land-use and land-cover (LULC) semantic segmentation, differing in modality, spatial resolution, geographic scope, and semantic granularity. Table 1 summarizes representative datasets but excludes aerial collections (e.g., LandCover.ai [4], MiniFrance [6], OpenEarthMap [31]) and RGB-only satellite data (e.g., LoveDA [28], WHDLD [22]), focusing instead on satellite sources that provide global coverage and include spectral and SAR information.

Optical multispectral datasets such as GID [26], LandCoverNet [1], Dynamic World [5], SeasoNet [12], Five-Billion-Pixels [27], and OpenSentinelMap [11] have established strong baselines for global LULC mapping. They leverage Sentinel-2 or Gaofen imagery, offering resolutions between 4 m and 10 m and between 5 and 33 classes. While they provide extensive optical coverage, their labels are often derived from pre-existing land-cover products or crowd-sourced sources, leading to spatial and temporal inconsistencies, especially in urban areas. Regional datasets such as WUSU [24] and ISASeg [8] achieve higher annotation

quality but lack geographic diversity. Agriculture-specific collections (e.g., Sen4AgriNet, PASTIS-R [19]) offer detailed crop-type classes but omit urban structures.

Multimodal datasets combine SAR with optical images (and other data modalities) to exploit complementary sensing capabilities. SEN12MS [21] remains a widely used benchmark, providing 181k global Sentinel-1/2 patches with 17 classes. However, its labels are derived from 500 m MODIS products, limiting reliability in fine-scale environments. reBEN [7] improves quality by enforcing label coverage thresholds ($> 75\%$) across 10 European countries. However, it is limited to Europe and its reference data is based on CORINE, i.e. a pre-existing semi-automatically created LULC product. Globe230k [23], MultiSenGE [30], and PASTIS-R [19] extend multimodal annotation coverage at varying spatial scales. Recent large-scale efforts such as MMEarth [16] and TerraMesh [3] further expand multimodal data to millions of samples with global extent, primarily targeting foundation-model pretraining rather than providing high-fidelity ground truth. Although invaluable for representation learning, their labels are largely derived from coarse or automated sources and thus unsuitable for precise supervised benchmarking.

Despite these advances, existing datasets face several limitations: (1) many provide only a single sensing modality; (2) several are thematically narrow, emphasizing specific domains such as agriculture or vegetation dynamics, limiting their applicability to the diverse land-cover compositions; (3) most rely on coarse external classification maps (e.g., MODIS, CORINE, ESA WorldCover), introducing temporal and spatial noise; and (4) nearly all are organized into small, predefined patches, limiting analysis of spatial context and urban morphology. As a result, current benchmarks are well-suited for large-scale pretraining but less appropriate for evaluating models on realistic, continuous urban scenes.

The proposed *50 Cities* dataset addresses these gaps by offering global, multimodal, and context-preserving coverage across six continents. It comprises 33 labeled and 17 unlabeled cities with co-registered Sentinel-1 (SAR) and Sentinel-2 (MSI) imagery at 10 m resolution. Unlike patch-based collections, *50 Cities* provides large, continuous urban scenes with surrounding land-cover types, enabling context-aware segmentation without artificial boundaries. The labeled subset includes 13 land-cover classes spanning diverse urban densities, industrial and vegetated areas, water, snow, sand, and bare soil. Rather than relying on existing low-resolution or (semi-)automated LULC products, all annotations are created manually to ensure high semantic accuracy and spatial consistency.

While smaller in labeled area than massive pretraining datasets, *50 Cities* uniquely combines manual high-accuracy labels, dual-modality sensing, and continuous ur-

Table 1. Comparison of major remote sensing datasets for LULC semantic segmentation. Our *50 Cities* dataset (highlighted) uniquely combines global urban coverage, multi-modal (SAR+MSI) imagery, and high-quality manual annotations.

Dataset	Year	MSI	SAR	Coverage	# Images/Patches	Patch/Img Size [px]	Res. [m/px]	# Classes
WUSU [24]	2023	✓	✗	China	2	$\approx 6,700 \times 5,700$	1	11
Ticino [2]	2024	✓	✗	Italy	1,502	256×362	5	8; 10
ISASeg [8]	2025	✓	✗	China	11,559	160×160	10	2
Five-Billion-Pixels [27]	2023	✓	✗	China	150	$6,800 \times 7,200$	4	24
GID [26]	2020	✓	✗	China	150	$6,800 \times 7,200$	4	5; 15
WHU-OHS [13]	2022	✓	✗	China	42 (7,795)	512×512	10	24
SeasonNet [12]	2022	✓	✗	Germany	1,759,830	120×120	10	33
Sen4AgriNet [25]	2022	✓	✗	France, Spain	225,000	—	10	168
LandCoverNet [1]	2020	✓	✗	Global	300 (9,000)	256×256	10	7
Dynamic World [5]	2022	✓	✗	Global	24,000	510×510	10	9
OpenSentinelMap [11]	2022	✓	✗	Global	137,045	192×192	10	15
PASTIS-R [19]	2022	✓	✓	France	2,433	128×128	10	18
C2Seg-BW [10]	2023	✓	✓	China	2	$\approx 10,000 \times 8,700$	10	13
C2Seg-AB [10]	2023	✓	✓	Germany	2	$\approx 840 \times 2,000$	10	13
WHU-OPT-SAR [14]	2022	✓	✓	China	100	$5,556 \times 3,704$	2; 5	7
MultiSenGE [30]	2022	✓	✓	France	8,157	256×256	10	14
reBEN [7]	2025	✓	✓	Europe	549,488	120×120	10	19
Globe230k [23]	2023	✓	✓	Global	6,198	512×512	1.0	10
MMEarth [16]	2025	✓	✓	Global	1,200,000	128×128	10	9/11
SEN12MS [21]	2019	✓	✓	Global	180,662	256×256	10	17
SSL4EO-S12 [29]	2023	✓	✓	Global	250,000	264×264	10	—
TerraMesh [3]	2025	✓	✓	Global	9,000,000	264×264	10	9
50 Cities	2026	✓	✓	Global	33 (+17)	$\approx 3,500 \times 3,500$	10	13

ban coverage. It thereby serves as a robust benchmark for evaluating semantic segmentation, multimodal fusion, and domain-generalization methods in global urban environments.

3. The 50 Cities Dataset

The proposed dataset comprises 50 co-registered Sentinel-1 and Sentinel-2 image pairs of metropolitan regions across six continents, each covering a major city and its surrounding areas. Among these, 33 scenes include manually annotated semantic label maps describing the land-cover and land-use characteristics, forming the labeled subset for supervised training and evaluation. The remaining 17 unlabeled scenes provide only the satellite imagery and are intended for self-, semi-, or unsupervised learning methods that leverage intrinsic data representations.

3.1. Data Collection

Satellite imagery was collected using the Copernicus Browser, enabling selection by location, satellite mission, product type, and cloud coverage. For this dataset, Sentinel-1 (S1) Level-1 SLC and Sentinel-2 (S2) Level-2A products were chosen. The S1 product provides complex radar data with full phase information, supporting precise radiometric calibration and geometric correction, while S2 Level-2A offers atmospherically corrected surface reflectance data. To ensure data quality, S2 scenes were restricted to a maximum of 10% cloud coverage, with additional manual filtering of

cloudy areas during labeling. Temporal proximity between S1 and S2 acquisitions was prioritized to minimize land-cover inconsistencies.

Figure 1 illustrates the global distribution of the 33 labeled and 17 unlabeled cities. The selected locations of the labeled subset span six continents and a wide range of geographic, climatic, and urban conditions - from tropical to temperate zones, and from dense megacities to mixed natural-urban regions. This diversity supports the development and evaluation of segmentation models with strong cross-domain generalization capabilities.

The unlabeled subset extends coverage to additional regions, including Central and South America, equatorial Africa, Eastern Europe, Southeast Asia, and the Pacific Islands. These areas were intentionally selected to complement the labeled subset and enhance global and environmental diversity. By capturing varied urban morphologies, vegetation types, and climatic contexts, the *50 Cities* dataset provides a comprehensive foundation for supervised and semi-supervised multimodal land-cover segmentation.

3.2. Data Preprocessing and Alignment

All imagery was preprocessed using ESA’s Sentinel Application Platform (SNAP) to ensure consistency and reproducibility across scenes. For Sentinel-1 (S1) data, the workflow included orbit file application, radiometric calibration, debursting, multilooking, and terrain correction, followed by georeferencing and resampling to a uniform 10 m reso-

lution. These steps correct geometric distortions and standardize radar backscatter for quantitative analysis. Sentinel-2 (S2) imagery was similarly geocoded. The 20 m and 60 m spectral bands were upsampled to 10 m to ensure spatial alignment across all channels and with the S1 data.

After preprocessing, both S1 and S2 scenes were cropped to a shared region of interest (ROI) centered on the main urban area of each city while extending beyond administrative boundaries to include diverse surrounding land-cover types such as vegetation, water, and infrastructure. This design supports balanced class distributions and avoids overrepresentation of homogeneous land-cover categories like forest or agriculture.

The dataset provides both the original raw products allowing users to apply custom preprocessing pipelines and the fully preprocessed scenes stored as GeoTIFFs, accompanied by RGB composites for visual inspection.

3.3. Manual Semantic Annotation

The semantic label map of each scene (examples in Fig. 1) is created manually by visually examining the RGB Sentinel-2 imagery along with publicly available auxiliary sources such as Google Maps, Google Street View, Google Earth 3D topography, and online references. The following describes the 13 predefined classes used for annotation, along with their color codes and abbreviations:

- **Low vegetation, bare (LV-B):** Sparse, low-growing plants with visible ground, rocks, or soil (e.g., shrubland, patchy farmland).
- **Low vegetation, non-bare (LV-NB):** Dense, low plants with minimal exposed ground, such as cultivated fields or golf courses.
- **High vegetation, bare (HV-B):** Taller plants or trees with large gaps between them (e.g., savannas, partially cleared forests).
- **High vegetation, non-bare (HV-NB):** Dense high vegetation without gaps, typically forests.
- **Bare soil (BS):** Areas with little to no vegetation or rocks, e.g., uncultivated fields, construction sites, or surface mines.
- **Sand (S):** Pure sand, e.g., deserts or beaches.
- **Rock (R):** Solid rock formations or rocky terrain.
- **Snow and ice (SI):** Snow-covered or icy surfaces such as glaciers.
- **Water (W):** Liquid water in oceans, lakes, rivers, or reservoirs.
- **Industry (I):** Industrial or commercial areas including warehouses, ports, airports, and shopping malls.
- **Urban low density (ULD):** Sparse settlements such as villages or small suburbs.
- **Urban medium density (UMD):** Densely built suburbs or small-town centers with multi-story houses.
- **Urban high density (UHD):** City centers and areas

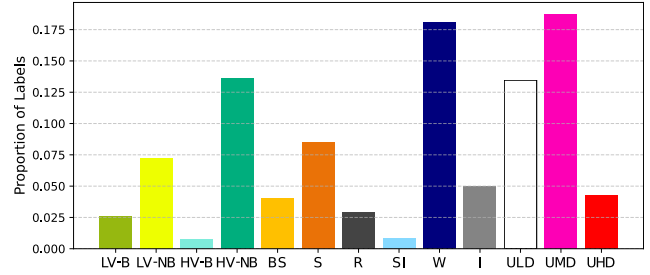


Figure 2. Pixel-wise distribution of the 13 semantic classes across all labeled cities (background excluded), showing that urban and water classes dominate, vegetation and geological classes are less frequent, and snow is rare.

with dense, multi-story or high-rise buildings.

Annotations aim for high pixel coverage while maintaining high confidence. Ambiguous or unknown regions are left unlabeled (background class). Each labeled scene was initially annotated by one annotator and subsequently reviewed, checked, and corrected by five additional annotators, resulting in a total of six independent assessments per scene. Quality assessment and control protocols were applied throughout this process to ensure consistency, minimize label noise, and maximize the reliability of the dataset. This multi-stage, collaborative annotation process is a key strength of the *50 Cities* dataset, providing high-accuracy, manually curated semantic maps suitable for rigorous benchmarking and model development.

3.4. Dataset Statistics

The annotation process described in Section 3.3 produced high-quality label maps covering nearly 36,000 km², with an average pixel coverage of 75.2% (minimum 49.7% for Shanghai, maximum 96% for Kyoto), yielding more than 270 million labeled pixels.

Fig. 2 shows the overall label distribution¹. Urban medium-density areas constitute the largest single class (18.7%), while all urban classes together (industry, low/medium/high density) cover 41.5% of labeled pixels. Water is the second most frequent class at 18.1%, and high vegetation non-bare accounts for 13.6%. Other vegetation classes collectively make up 24.2%, and geological classes (bare soil, sand, rock) represent 15.5% of labeled pixels. Snow is the least frequent class (0.8%).

Fig. 3 presents a UMAP [15] visualization of the stacked Sentinel-1 (2 channels) and Sentinel-2 (12 channels) data, color-coded by semantic class (Fig. 3a) or continent (Fig. 3b)². The projection forms a dominant cluster with minor offshoots, indicating continuity of spectral-backscatter

¹Per-continent distributions are provided in the supplemental material.

²Class-wise and continent-wise projections are provided in the supplemental material.

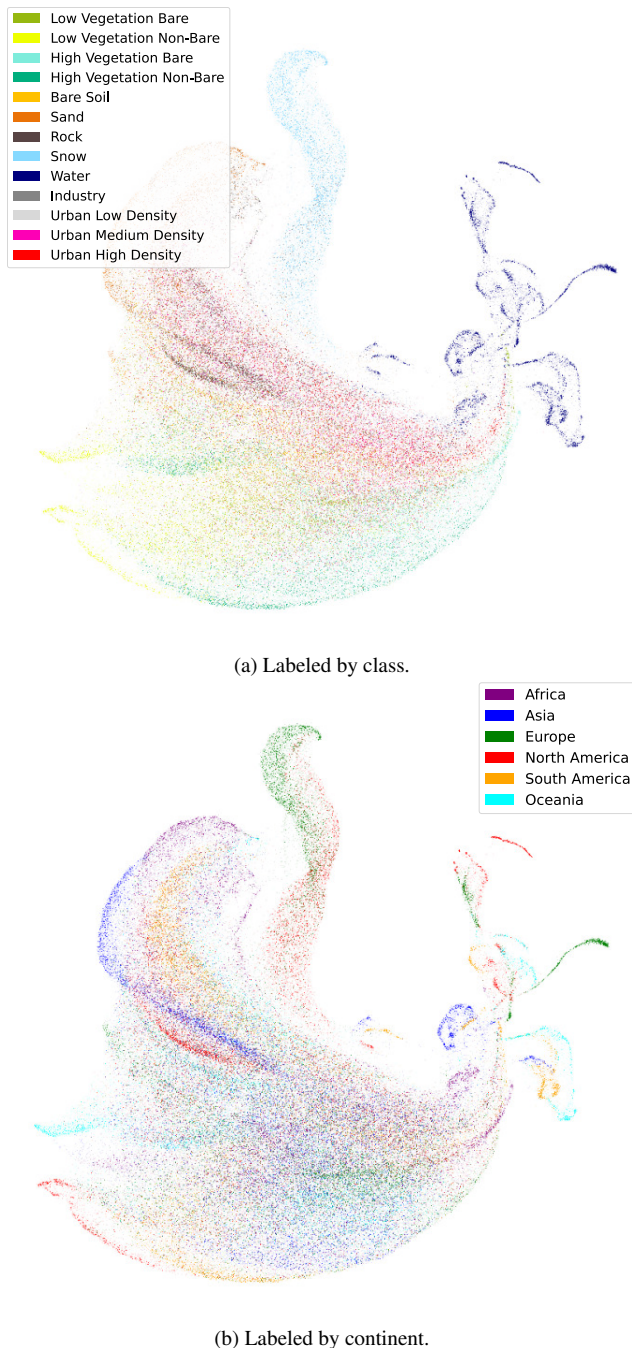


Figure 3. UMAP embedding of all pixels from the *50 Cities* dataset, illustrating that the 14-dimensional Sentinel-1+Sentinel-2 feature space captures meaningful semantic separation (a) while also revealing geographic patterns and subtle domain shifts between continents (b).

characteristics across land-cover types. Within the main cluster, distinct regions correspond to semantic classes: vegetation classes form coherent subgroups reflecting spectral similarity, urban areas of different densities cluster to-

gether with less separation, and sand and rock occupy compact, well-defined regions. Bare soil overlaps with sparsely vegetated areas, while water and snow form isolated clusters; snow is compact, whereas water shows fragmentation likely due to regional and sensor variability.

When colored by continent, the UMAP projection reveals considerable intermixing of samples, indicating that similar land-cover types share comparable spectral and backscatter properties globally. However, some regional patterns persist: for example, sand is primarily from Africa, Asia, and South America, and snow originates mostly from Europe and North America. These patterns demonstrate that, while the feature space is broadly coherent, subtle domain shifts exist between continents.

Overall, the UMAP results highlight two key properties of the dataset. First, the stacked Sentinel-1 and Sentinel-2 feature space effectively captures semantic structure across diverse land-cover types. Second, the dataset reflects real-world geographic and regional variations, exposing models to domain differences they are likely to encounter in practice. Together, these characteristics make *50 Cities* a valuable resource for developing robust semantic segmentation methods with global generalization capabilities.

4. Experimental Evaluation

To evaluate the utility and real-world relevance of the *50 Cities* dataset, we design a series of experiments that examine both baseline performance and cross-domain robustness. We employ two complementary segmentation architectures - UNet [18], a convolutional encoder-decoder effective for fine-grained spatial details, and SegFormer [32], a transformer-based model capturing long-range dependencies and global context. Section 4.1 establishes baseline results on the full dataset, while Section 4.2 investigates cross-continental generalization, testing models trained on one region against others to assess geographic transferability. Section 4.3 isolates the contribution of each sensor modality by training on Sentinel-1 and Sentinel-2 data separately. Finally, Section 4.4 explores the benefit of unlabeled data through semi- and self-supervised learning, as well as fine-tuning a geospatial foundation model. All results are reported in mean Intersection-over-Union (mIoU)³.

4.1. Full-Dataset Baselines

To establish baseline performance on the labeled portion of the dataset, U-Net [18] and SegFormer [32] are employed. In this first experiment, both models are trained on patches of stacked S1 and S2 from all 33 labeled cities, with 20% of the image tiles being held out for testing. This setup provides insight into the overall difficulty of the benchmark

³Hyperparameters and model sizes are provided in the supplementary material.

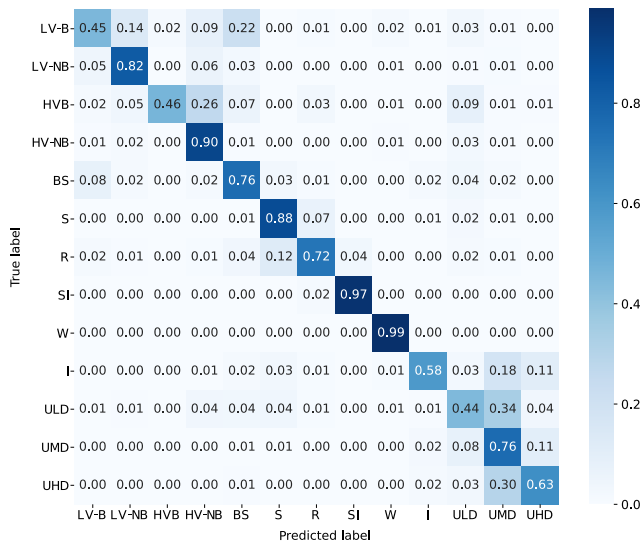


Figure 4. Confusion matrix of the SegFormer model trained on all labeled cities using Sentinel-1+2 inputs. The matrix illustrates class-wise segmentation performance across the 13 land-cover categories, highlighting strong accuracy for vegetation and water classes, and partial confusion among urban density levels.

and assesses how well standard architectures can generalize across geographically diverse urban areas and surrounding environments.

Both models achieve solid performance on the full dataset, with U-Net reaching an mIoU of 64.2 and SegFormer achieving 57.8. A class-wise analysis of the SegFormer results (Fig. 4) shows that performance varies strongly by land-cover type. High vegetation (non-bare) and water achieve the highest accuracies (0.90 and 0.99, respectively), indicating that large, spectrally distinct classes are well captured. Similarly, snow and sand are reliably identified, though with minor confusion with visually similar geological surfaces. Urban classes are generally well separated by density level, although some confusion persists between low-, medium-, and high-density areas. Industrial regions show notable overlap with medium- and high-density urban areas, reflecting their spectral and structural similarity. Among vegetation types, low and high vegetation - especially bare variants - are the most frequently confused, highlighting the challenge of discriminating mixed or sparsely vegetated terrain.

These findings illustrate that while modern architectures capture major land-cover patterns effectively, they still struggle with subtle distinctions in heterogeneous or mixed-use areas, leaving substantial headroom for future improvements. Neither model achieves near-perfect segmentation, suggesting that the dataset can support a wide range of performance outcomes, making it a valuable benchmark for evaluating new methods.

4.2. Testing Across Continents

The second experiment evaluates cross-continental generalization using six data splits, one for each continent represented in the dataset. In each run, all tiles (consisting of stacked S1 and S2 images) from a single continent are used for training, with 20% randomly sampled for validation and the remaining 80% forming the training set. Data from the other continents serve as independent test sets, allowing assessment of how well models generalize to geographically distinct regions. This setup highlights intercontinental differences in land-cover characteristics and compares the models' ability to learn from limited, potentially unbalanced regional data.

The results in Table 2 show that both U-Net and SegFormer achieve the highest mIoU when evaluated on the same continent on which they were trained. For example, U-Net reaches 52.2 mIoU on African test patches when trained on Africa and 62.3 on South America when trained on South America. SegFormer follows the same trend but consistently yields slightly lower scores. These results confirm that segmentation performance is strongly tied to the similarity between training and test domains, reflecting realistic geographic and environmental dependencies in land-cover and urban structure.

Cross-continental performance is substantially lower, highlighting the challenges of domain shift between regions. For instance, a model trained on Africa achieves only 15.2 mIoU on Asia and 24.5 on Europe, while U-Net trained on Europe achieves 15.0 on Africa and 18.0 on Asia. This pattern is consistent across all continents and both architectures, confirming that differences in land-cover distributions and spectral-radar characteristics across continents strongly affect segmentation performance. Despite these drops, some transfer remains possible. For example, U-Net trained on South America still achieves 32.5 mIoU on Asian test data, indicating that certain features are shared globally, even if overall generalization is limited.

On average across all off-diagonal entries, U-Net retains 22.7 mIoU and SegFormer 17.6, confirming that both architectures experience a substantial performance reduction when applied to unseen continents. Among all regions, Asia exhibits the strongest cross-domain transfer, with an average mIoU of 25.2 across other continents for U-Net and 21.2 for SegFormer. This generalization likely arises from Asia's high internal diversity, covering tropical, arid, temperate, and mountainous climates. In contrast, models trained on more homogeneous regions such as Europe or North America achieve lower mean transfer scores, indicating narrower environmental coverage.

Overall, the results reveal that models trained on one continent perform best within that domain and degrade significantly elsewhere. This behavior underscores both the realism and the challenge of the *50 Cities* dataset: it cap-

Table 2. Cross-continental generalization performance (mIoU) for U-Net and SegFormer (trained on stacked S1 + S2 data). Each row indicates the training continent and each column the test continent. Bold values mark in-domain (same-continent) performance. Both models perform best on their home regions but show substantial performance drops across continents, reflecting realistic domain shifts in global-scale segmentation.

	Africa		Asia		Europe		North America		Oceania		South America	
Training	U-Net	SegFormer	U-Net	SegFormer	U-Net	SegFormer	U-Net	SegFormer	U-Net	SegFormer	U-Net	SegFormer
Africa	52.2	35.4	15.2	10.3	24.5	9.9	21.9	13.6	25.0	10.9	23.6	12.0
Asia	24.7	20.6	58.8	48.0	24.1	22.4	21.8	13.8	30.5	21.8	25.0	27.5
Europe	15.0	11.8	18.0	16.9	56.7	57.8	22.7	14.6	13.1	24.9	19.2	17.5
North America	13.0	14.7	17.1	14.3	27.2	26.6	50.8	46.7	23.9	23.4	18.9	19.1
Oceania	22.6	10.6	22.5	14.8	29.3	16.1	22.8	12.7	60.7	44.4	24.7	20.8
South America	25.8	21.7	32.5	22.4	26.7	23.6	18.9	15.0	30.0	22.5	62.3	51.8

tures authentic geographic diversity and thus provides a robust testbed for developing segmentation methods with improved cross-continental generalization.

4.3. Sensor-Specific Performance

This experiment evaluates how different input modalities affect segmentation performance. Using the Asia subset, which includes cities from the Middle East, India, and East Asia, U-Net and SegFormer are trained on Sentinel-1, Sentinel-2, and their combination. Asia is chosen for its broad diversity in climate, land-cover types, and urban morphology, providing a representative testbed for assessing the individual and joint contributions of radar and optical data. This setup enables a controlled comparison of Sentinel-1, Sentinel-2, and multimodal inputs to quantify the benefits of data fusion in semantic segmentation.

Results in Tab. 3 for the Asia test set show a clear dependence of performance on the chosen modality. Models trained solely on Sentinel-2 data achieve substantially higher mIoU than those trained on Sentinel-1, with U-Net reaching 56.4 and SegFormer 47.2, compared to 42.1 and 20.5, respectively, for Sentinel-1. This confirms the dominant role of optical multispectral information in distinguishing between vegetation, urban, and natural classes, whereas radar backscatter alone provides limited class separability, particularly for the transformer-based SegFormer. Combining Sentinel-1 and Sentinel-2 data yields the best overall results for both architectures, with U-Net reaching 58.8 mIoU and SegFormer 48.0. The improvement over Sentinel-2 alone is modest but consistent, suggesting that radar data contributes complementary structural and textural cues that enhance optical-based segmentation.

Cross-continental evaluation further supports this trend: Sentinel-2 consistently outperforms Sentinel-1, while the fused configuration provides small yet reliable gains across all regions. Overall, these findings highlight that while multispectral imagery remains the strongest single modality for LULC segmentation, the integration of SAR data enhances robustness and generalization, particularly in heterogeneous or cloud-prone environments.

4.4. Learning with Unlabeled Data

This experiment evaluates label-efficient learning strategies that exploit the unlabeled portion of the dataset. Three approaches are considered: a semi-supervised U-Net (SemiSL), a self-supervised model (SSL), and a geospatial foundation model (GFM).

In the semi-supervised setting, a U-Net is first trained on the labeled data of Asia to generate pseudo-labels for the unlabeled subset, keeping only pixels with confidence above 90%. This allows the model to leverage additional data without further manual annotation. In the self-supervised approach, following Scheibenreif et al. [20], the model backbone is pretrained on both labeled and unlabeled data using a contrastive objective that aligns Sentinel-1 and Sentinel-2 embeddings, after which a segmentation head is trained on the labeled subset of Asia.

To complement these models, we evaluate CROMA [9], a recently proposed geospatial foundation model pretrained on the large-scale SSL4EO-S12 dataset. CROMA learns general multimodal representations from globally distributed Sentinel-1 and Sentinel-2 imagery. For this experiment, a lightweight segmentation head is attached to its frozen backbone and fine-tuned on the labeled portion of Asia in the *50 Cities* dataset. This setup tests how well broad, globally pretrained features transfer to the fine-grained, urban-scale segmentation task defined here.

The results in Tab. 4 show that compared to the fully supervised baseline (58.8 mIoU on Asia), the semi-supervised model achieves a slightly higher 59.4, indicating that pseudo-labeling effectively extends the training set and improves generalization within the same domain. The self-supervised model reaches 41.5 mIoU, lower in absolute accuracy but showing more consistent cross-continental performance, suggesting that contrastive pretraining yields more domain-invariant but less class-specific features. The foundation model (CROMA) achieves 28.4 mIoU on Asia - below the supervised baselines but notable given that it was pretrained on a different global dataset without task-specific labels. Its relatively balanced performance across

Table 3. Performance (mIoU) for U-Net and SegFormer trained on the Asia subset using different input modalities (Sentinel-1, Sentinel-2, and their combination). Optical Sentinel-2 data yield the highest single-modality performance, while multimodal fusion (S1+S2) provides consistent but modest improvements across all continents.

	Africa		Asia		Europe		North America		Oceania		South America	
Sensors	U-Net	SegFormer	U-Net	SegFormer	U-Net	SegFormer	U-Net	SegFormer	U-Net	SegFormer	U-Net	SegFormer
S1	12.3	3.6	42.1	20.5	21.1	7.4	9.6	6.7	16.8	7.4	21.4	7.9
S2	21.8	19.7	56.4	47.2	23.3	21.0	20.8	13.1	32.8	21.9	27.2	22.0
S1+S2	24.7	20.6	58.8	48.0	24.1	22.4	21.8	13.8	30.5	21.8	25.0	27.5

Table 4. Performance (mIoU) of models leveraging unlabeled data. Results are shown for semi-supervised (SemiSL), self-supervised (SSL), and geospatial foundation model (CROMA) approaches trained on the Asia subset and evaluated across all continents.

	Africa	Asia	Europe	N. America	Oceania	S. America
SemiSL	23.7	59.4	23.6	21.2	30.6	23.3
SSL	10.1	41.5	15.6	16.7	18.3	23.2
CROMA	13.7	28.4	16.7	14.8	21.3	14.8

continents demonstrates good out-of-domain generalization and highlights the potential of large-scale pretraining for global transfer.

Overall, these results illustrate a clear hierarchy of label efficiency: semi-supervised learning provides modest but tangible gains, self-supervised learning improves domain robustness, and geospatial foundation models offer promising transferability across unseen regions. Together, they confirm that the *50 Cities* dataset is a suitable benchmark for advancing both supervised and label-efficient learning in global multimodal remote sensing.

5. Conclusion and Limitations

This work introduces *50 Cities*, a globally distributed benchmark for multimodal semantic segmentation in Earth observation. The dataset provides co-registered Sentinel-1 (SAR) and Sentinel-2 (MSI) imagery for 50 metropolitan regions across six continents, including 33 cities with manually annotated, pixel-accurate LULC maps and 17 unlabeled cities for semi- and self-supervised learning. Unlike patch-based datasets, *50 Cities* offers large, continuous urban scenes with surrounding landscapes, enabling context-aware segmentation and reducing sampling bias. Each sample includes both original Level-1/Level-2 data and fully preprocessed, georeferenced imagery, ensuring reproducibility and flexibility for diverse research workflows.

Comprehensive experiments with CNN-, Transformer, and label-efficient models demonstrate that the dataset supports a wide range of segmentation approaches and provides a challenging yet realistic benchmark for assessing generalization. While models achieve solid within-region performance, cross-continental experiments reveal pronounced

domain shifts, underscoring the difficulty of transferring models trained on geographically limited datasets. This highlights a key insight: even large regional datasets cannot guarantee global generalization, emphasizing the need for globally diverse and manually curated benchmarks such as *50 Cities*.

Despite its global scope, *50 Cities* has several limitations. First, Southeast and Northern Asia, as well as Pacific island nations such as Fiji and Polynesia, are underrepresented in the labeled subset. Second, the dataset is temporally static, capturing each city at a single point in time without accounting for seasonal or interannual variation. Third, while it provides high annotation quality, its total size remains moderate compared to very large pretraining corpora such as MMEarth or TerraMesh, which however rely on coarse, automatically generated maps rather than manual reference labels. These limitations point to future opportunities for expanding both spatial and temporal diversity while maintaining annotation fidelity.

Future work will focus on extending the dataset temporally, incorporating multi-season and multi-year imagery to enable dynamic land-cover monitoring. Additional urban areas and unlabeled regions will be included to further enhance coverage, particularly across underrepresented continents and climate zones. The dataset also opens avenues for systematic benchmarking of advanced segmentation methods, including state-of-the-art supervised architectures, self-supervised pretraining schemes, and geospatial foundation models. By enabling robust, reproducible, and globally representative evaluation, the *50 Cities* dataset lays the foundation for the next generation of scalable Earth observation models.

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